The Hong Kong Polytechnic University

Subject Description Form

Subject Code	ABCT3115
Subject Title	Biochemical Techniques Laboratory
Credit Value	1
Level	3
Pre-requisite/ ~~Co-requisite/~~ ~~Exclusion~~	General Laboratory Techniques and Safety, Biochemistry
Objectives	This subject is intended to apply fundamental principles of instruments and techniques commonly used in biochemistry and molecular biology through experimentation.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Demonstrate proficiency in a range of biochemical laboratory techniques, including chromatography, electrophoresis, and spectrophotometry, and to apply these techniques to analyze and interpret experimental data. b. Develop the ability to design and conduct experiments, including formulating hypothesis, selecting appropriate methodologies, and anlyzing results, to investigate biochemical processes and phenomena. c. Students will acquire skills in quantitative data analysis and interpretation through written lab reports.
Subject Synopsis/ Indicative Syllabus	• Principles & practices of chromatography – permeation, ion exchange, affinity chromatography; gas chromatography; HPLC; separation of biomolecules • Electrophoresis: principles and general techniques; special techniques-high voltage discontinuous, SDS-polyacrylamide, gradient, isoelectric focusing, two-dimensional electrophoresis and western blotting • Fluorescence principle and its applications; flow cytometry • DNA microarray • DNA sequencing and fingerprinting • Luminescence and chemiluminescence; luciferase, green fluorescent protein • Real-time PCR

- Sample preparation: cell disruption, buffer, dialysis, concentration methods, measurement of proteins and nucleic acids
- Centrifugation and subcellular fractionation

Teaching/Learning Methodology

Practicals will be used to allow students to have a hands-on experience in purifying and analyzing proteins. Lab reports are used to teach proper write up and communication skills.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c		
1. Lab attendance	10%	✓	✓			
2. Lab reports	90%	✓	✓	✓		
Total	100%					

Students are required to submit laboratory reports which include the explanation of the principles of the techniques used. But more importantly, we require students to present the experimental results in a clear and organized manner and be able to interpret the results.

Student Study Effort Expected

Class contact:	
▪ Practicals	24 Hrs.
Other student study effort:	
▪ Laboratory Preparation (reading manuals and related background materials)	3 Hrs.
▪ Report writing and data analysis	24 Hrs.
Total student study effort	51 Hrs.

Reading List and References

Text book:

Statistics. Donnelly RA Jr and Abdel-Raouf F. 2016. Alpha, Indianapolis, Indiana. ISBN : 9781465454096

Recommended reading:

The lady tasting tea : how statistics revolutionized science in the twentieth century. Salsburg D. 2001. Freeman, New York, New York. ISBN : 0716741067

Library location: Q175.S2345 2001

Hands-on exploratory data analysis with R: Become an expert in exploratory data analysis using R packages. Datar R, Garg H. 2019. Packt Publishing, Birmingham, UK. ISBN: 178980437X

Library online access